The Great Rewiring

Two Resource Grabs, One Network: Pricing Modern Mercantilism and AI on the World Trade Web

William E. Shields

August 1, 2026

Abstract

Bridgewater has named the present chapter two resource grabs: a mercantilist grab for chokepoints and an AI grab for compute [1]. We price both against a single object, the world trade network, and find one process rather than two. Priced separately, the two understate each other, because the compute grab's physical inputs cross exactly the chokepoints the mercantilist grab is weaponizing.

For roughly thirty years, nearly every country picked its trade routes the same way, on price. Then countries discovered that depending on a rival for something essential can be used against them, through export bans, blacklists, and mineral controls. So the question flipped from which route is cheapest to which route is safest, and a trading system built for safety wires itself differently from one built for cost. Artificial intelligence pours fuel on the flip: the race to build it runs on chips, minerals, and grid equipment, exactly the goods being weaponized, so the network's most valuable cargo now moves along its most contested routes. We call that rewiring the Great Rewiring, we measure it on the actual map of who trades what with whom, and we make ten specific predictions about how far it runs by 2028. Every prediction carries a probability and a named public data source that will settle it, so anyone can score us later.

For twenty-five years, one number explained the shape of world trade. A published model fed nothing but each country's GDP reproduced who traded with whom, and through which hubs. We hold that machine fixed and run it forward 300 times. That gives a null distribution for a world that never rewired. The present reads as deviations from it. The model still knows everyone's size, and on the ordinary map it is still right. Its misses concentrate at the top of the size distribution, in the corridors and links policy now names. That is how we know the wiring changed rather than the weather.

Two benchmarks bracket the deviations. Against a size-only gravity benchmark refitted each year, China–US now carries approximately 0.35× the trade its economic size implies, and Vietnam–US 16.2×. The no-rewiring generator, measuring only movement since 2020, puts China–US at approximately 0.86× its benchmark median, near the 34th percentile of comparable country pairs.

Our sharpest number did not survive its own audit. A placebo test broke our own inference chain, and the corrected reading prints smaller than our first draft claimed. The topology of world trade refuses to fragment under attack, yet strength-targeted removal strands roughly ten times the trade value random failure does.

Base rates were re-priced the day an audit found one broken, and deliberate bets are labeled as bets.

We forecast that this re-pricing is a change of objective, not the turn of a cycle, that it continues through 2028, and that any framework mean-reverting to the pre-2020 network is mispricing it. The slate prices how fast and in what order. It also names what would prove us wrong.

Replication: Generator, analysis scripts, processed data, and an agent-readable capability map at https://github.com/Synthminds/GreatRewire.

Declaration of AI use. During the preparation of this work we used ChatGPT, Claude, Perplexity, and GPAI. We used them to expand the option space, generating alternative framings and counterarguments for us to evaluate. We used them to surface candidate patterns across studies and datasets, each of which we then checked against the sources ourselves. We also used them to challenge our assumptions and to hunt for gaps and bias in our own reasoning, and to improve readability. They assisted in writing and debugging the Python that builds the panel, runs the counterfactual ensembles, and produces the exhibits. We verified every output, result, and line of code against the underlying data and against established methods before it entered this paper. No probability, threshold, resolution criterion, or conclusion here was set by a model. We reviewed and edited all content, we take full responsibility for it, and the ideas put forward are our own.

Part 1: Ten Forecasts, Four Layers

GDP alone predicted a country's position in the world trade network until 2020. We measure the gap between what GDP predicts and what actually trades, then forecast how far it widens through 2028 and in what order. Priced apart, Bridgewater's two grabs [1] understate each other, and the compounding sits where the strategic layers overlap.

The forecasts are grouped by how fast each part of the network can answer the same shock. Governments move first, deleting and re-pricing trade routes within quarters (Group A). Trade flows re-route behind them, over quarters to years (Group B). The AI buildout answers on construction time, because transformers and transmission take years to build (Group C). Labor moves last (Group D).

Each forecast is one line, giving the probability, the claim, its deadline, and a **Reads** label naming the network behavior it tests. Appendix A names, for each of the ten, the published series that settles it, chosen so no judgment is required on the resolution date. Probabilities are the final Monte Carlo values from Appendix D, adjusted where noted. Seven of the ten sit below even because most thresholds are deliberately set above where the world already stands. The slate prices continuation. It does not price the status quo.

Group A. The policy layer: states delete links

A1. 62%. The US average effective tariff rate exceeds 8% for calendar 2027. *Reads:* Tariffs survive their court losses.

A2. 38%. The Bureau of Industry and Security (BIS), the US export-control regulator, adds 100 or more China-based entities to its Entity List in final rules citing semiconductor, advanced-computing, or AI grounds between August 1, 2026 and December 31, 2027. *Reads:* Export-control listings resume once the truce lapses.

A3. 45%. China enforces its critical-mineral export controls against at least one foreign-made product, documented by December 31, 2027. *Reads:* Beijing starts policing its exports beyond its own borders, as Washington already does.

Group B. The physical layer: the graph answers at two speeds

Flows re-route in quarters. Severed links either re-heal over years or do not.

B4. 45%. The combined share of US goods imports held by Mexico and ASEAN, the Association of Southeast Asian Nations, exceeds China's share by at least 30 percentage points in calendar 2027. *Reads:* Trade re-routes through the connectors faster than it returns to China. The threshold sits above the +25.7pp already measured year-to-date, so this prices continuation rather than the status quo.

B5. 20%. China and Hong Kong together account for 12% or more of NVIDIA's total revenue in fiscal 2028. *Reads:* The largest severed trade link does not grow back. A YES resolution means it did.

Group C. The AI layer: the grab meets the constraint

C6. 45%. Combined calendar-2027 capital expenditure of Microsoft, Alphabet, Amazon, and Meta, including finance leases, exceeds $1.0 trillion. *Reads:* AI spending keeps accelerating before it reaches its physical limit.

C7. 75%. US data-center electricity consumption exceeds 8% of total US generation in calendar 2028. *Reads:* Data centers take a large enough share of American electricity to bind on supply.

C8. 55%. At least 40 US states have approved data-center or large-load electricity tariffs by December 31, 2027. *Reads:* States write new electricity rules where the strain is worst.

C9. 42%. Year-over-year growth in combined Big-4 capital expenditure falls below 10% in two consecutive quarters before December 31, 2028, on the same basis as C6. *Reads:* AI spending growth stalls because power and grid capacity cannot keep pace.

C6 and C9 are one world, not two. The pair is not a bet on acceleration and deceleration at once. C6 asks whether spending clears $1.0 trillion in 2027; C9 asks whether its *growth rate* stalls at any point before end-2028. Spending can compound past a trillion and then decelerate, and on most Monte Carlo paths it does exactly that. The two forecasts price different moments on the same curve.

Group D. The slowest layer: the regime reaches the people who built it

D10. 35%. The US unemployment rate for computer and mathematical occupations exceeds the all-college-graduate rate by 1.5 percentage points or more in any calendar quarter before end-2027. *Reads:* The buildout's own labor market weakens first. Any tech-labor downturn resolves this, so it is the slate's loosest mechanism link, and we print it as one.

Slate balance. Conviction concentrates where the network mathematics is strong, at C7 and C8 and A1. It falls to near coin-flips where litigation and politics decide. We re-priced A2 downward the day an audit found its base rate frozen. Two further candidates we cut rather than print, one lacking a published series, the other reaching us only through a secondary source. A forecast that cannot be adjudicated cleanly is not a forecast.

Part 2: Framework, "The Great Rewiring"

For twenty-five years one variable, GDP, generated the world trade network's entire topology: degree structure, clustering, cores [2; lineage: 3–5]. Run that model past 2020 and the residual against the actual network grows. Corridors sit suppressed below the weight their economies imply, links die faster than the deletion law expects, and flows re-route through connectors. The Great Rewiring is what GDP can no longer explain (Exhibit X1-L).

Two interlinked resource grabs [1, 6] are, in network terms, one rewiring: the mercantilist grab re-prices and deletes links, the AI grab re-values nodes toward compute and power, and the two compound where the strategic layers overlap. Six sections price that flip.

§1. The objective function flipped

Centrality became coercive power [8], and every dependent country's optimum shifted from minimizing cost to minimizing dependency. As a program, the network moved from minimizing landed cost subject to a capacity constraint to minimizing dependency subject to a cost budget. That is a different optimization, not a worse solution to the old one, and it is why the shadow prices in Appendix C, each the price a binding constraint charges, are the right object to forecast against. The dual puts a price on what a chokepoint is worth to whoever controls it. The objective is not new. It is the military logistician's. A network designed against an adversary who interdicts it optimizes survivability and pays the efficiency premium deliberately. The civilian system is adopting it because commercial networks turned out to be interdictable too.

The regime survives its legal defeats. The tariff wall has been struck down twice and rebuilt within days both times under a different legal authority, most recently a Section 301 replacement covering 60 economies that took effect the minute its predecessor lapsed (Appendix A). A wall rebuilt twice under legal fire is a commitment, and A1 prices it as one. The convergence runs both directions, since China's extraterritorial mineral rules are Washington's own instruments, adopted by Beijing (Appendix A).

We know the price. Severe fragmentation costs up to approximately 7% of output, 8–12% with technology decoupling [9], and it is already being paid. Since 2022 cross-bloc trade and investment run approximately 12% and 20% below their within-bloc trends [10]. States pay selectively, highest-centrality links first, the way a planner armors the critical line of communication before the convenient one. This extends "We're All Mercantilists Now" [11] from narrative to objective function. *Anchors: A1–A3.*

§2. AI is both cargo and current

The AI grab becomes a physical claim where it meets the grid. The buildout, "the biggest capex buildout since World War II" [1], is worth roughly 100 to 140 basis points of 2026 US growth [12, 13] on approximately $725B of guided Big-4 capital expenditure.

That grab meets a constraint. Training compute grows roughly 4–5× per year toward multi-gigawatt power draws [14, 15], while data centers climbed from approximately 4.4% of US electricity in 2023 to 4.7% in 2024, with Lawrence Berkeley's June 2026 update putting the 2028 reference case at roughly 464 TWh [16, 17]. The binding margin is not generation. It is interconnection, the physical work of attaching new load to the grid. Transformer lead times run approximately 143 weeks against China's 48, and US ultra-high-voltage transmission stands at zero miles built against more than 30,000 miles in China [18]. Microsoft's contracted backlog reached $678B on June 30, 2026, while its executives describe holding chips they cannot yet plug in (Appendix C).

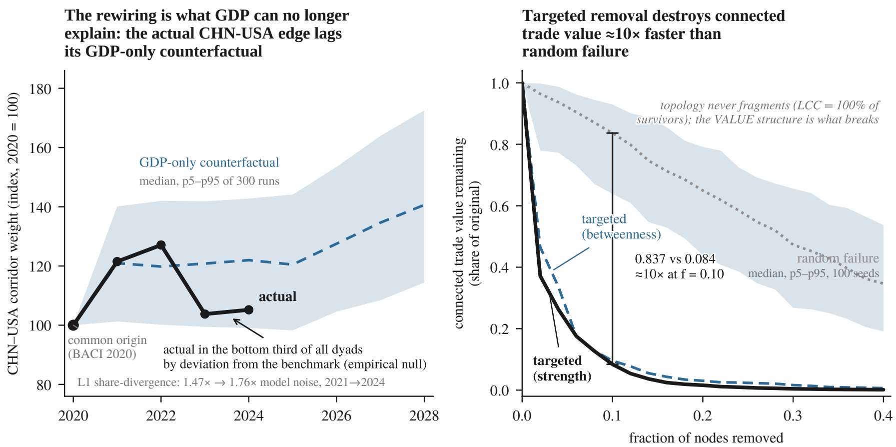
This buildout gets priced as capital [1, 12, 13]. The capital is real, but it is the wrong variable to forecast against. Money can be raised in a quarter; a transformer takes 143 weeks, and the queue clears at roughly 13% over a wait above five years. The binding input is delivery, not funding, which is why C7 and C8 carry more conviction than C6, and why C9 prices a stall no capital-availability model would generate. A forecast built on capex intentions will be early; one built on the queue will be late but closer. *Anchors: C6–C9.*

§3. Fragmentation is nonlinear: the network holds, then partitions

We run targeted-removal percolation on the 2024 network, one node at a time. Trade is a complex network in which shocks propagate with centrality and interconnection is robust-yet-fragile [3, 20, 21]; our simulation sharpens that law. Topological persistence at this density is a property of the object, so we treat it as the control. The giant component retains essentially all surviving nodes even at 30% targeted removal. The finding is the value structure. At approximately 10% strength-targeted removal, only 8.4% of original trade weight survives connected. Random removal at the same depth leaves roughly 83.7%. The gap is approximately 10×, most of it throughput the removed hubs carried themselves, and betweenness targeting collapses the structure the same way (Exhibit X1-R; Appendix B2).

The mechanism is a mismatch between two distributions. Degree is near-homogeneous at this density while edge weight is heavy-tailed, so removing the highest-strength nodes takes very little degree and most of the weight. Topological robustness and economic robustness are different properties of the same graph, and this network has one without the other (Appendix B2).

At the sector layer, single origins partition outright. NdFeB magnets run approximately 63% China, lithium-ion cells 59% China, memory chips 37% South Korea, wafer-fabrication equipment 26% Netherlands, logic chips 21% Taiwan (Appendix B1). Each is critical for the same reason. It sits upstream of something no state can quickly substitute. Magnets and cells sit beneath electric vehicles and precision weapons. Memory and logic chips sit beneath all computing, AI included, and wafer-fab equipment beneath the ability to make chips at all. Policy escalates where the constraints bind. Appendix C's shadow prices name the next moves. Two are already visible. One is the large-load tariff wave, now at 24 states. The other is the extraterritoriality buildout. *Anchors: A3, C8.*



Source: validated CompleNet-2022 port, 300-run ensemble (left); BACI 2024 percolation, 100-seed bands (right). Appendix B2. Computed 2026-07-30.

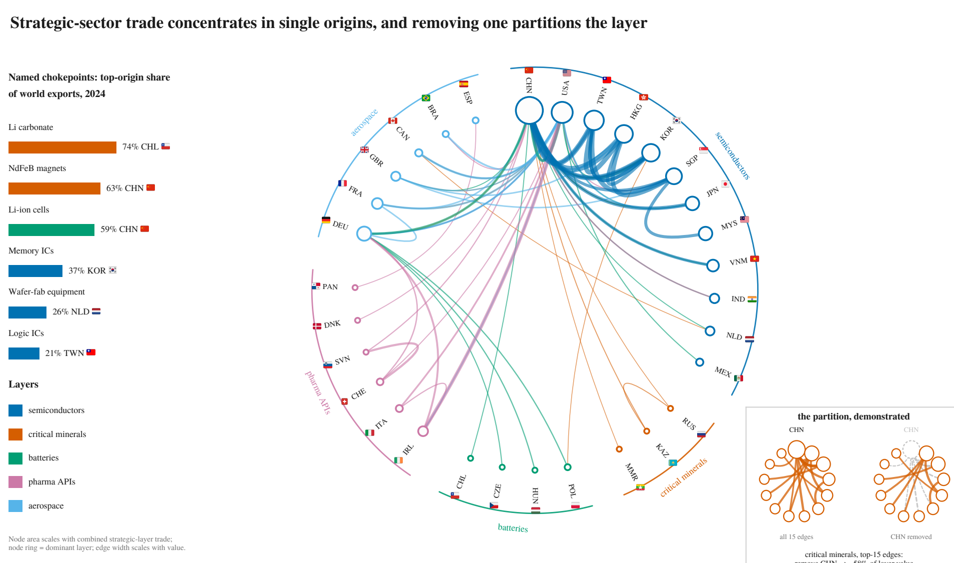
Exhibit X1. Left, the rewiring residual, rising, measured as actual against GDP-counterfactual divergence, 2021–2024 (Appendix B2). Right, targeted-attack percolation with Monte Carlo bands. The topology holds; the value structure does not.

§4. Who wins, who loses

Connectors are taking share. The Mexico-plus-ASEAN gap over China in US goods imports moved from roughly +11.6pp in 2023 to +25.7pp year-to-date 2026, and B4 re-arms at 30pp to price continuation. Vietnam–US runs at 16.2× its GDP-implied weight on the gravity benchmark and near the 75th percentile on the generator, the two readings agreeing in direction and differing in span (Appendix B2). Re-routing is not de-risking. The connectors' imports from China rise with their exports to the US [25, 26], and look-through exposure exceeds face-value trade [27], part of it Chinese content re-routed rather than replaced. B4 therefore prices flows, not independence.

Meanwhile the maximum-weight cross-bloc link is severed. China–US trades at approximately 0.35× its gravity-implied weight, down from roughly 0.48× in 2017, and near the bottom third on the generator (Appendix B3). It is enforced from both ends, Washington licensing the chips while Beijing caps purchases. B5 bets against re-healing, with the caution that billing location can move without end-use moving.

Power-rich regions clear the binding constraint (C7–C9). The slowest layer turns inward, and the network re-prices its own builders first. That is what D10 tests.



Source: CEPII BACI HS17 V202601, 2024; five strategic layers, top-15 edges each; layers and HHI per Appendix B1. Computed 2026-07-31. Flag icons: Twemoji, CC BY 4.0.

Exhibit X2. Five strategic layers of the 2024 trade web: nodes sized by strategic-layer trade, edge width by export value, where a single origin holds roughly 21–74% of world exports in each named chokepoint, from lithium carbonate (74%, Chile) to logic chips (21%, Taiwan), five of which Appendix B1 tabulates. Inset: removing the top origin strands approximately 58% of the critical-minerals layer's top-edge value (Appendix B1).

§5. The test that broke the chain

Our sharpest number did not survive its own audit. The inference above ran as follows: actual China–US weight sits near the 12th percentile of a 300-run no-rewiring ensemble, so the corridor has been rewired. That holds only if the ensemble is a calibrated null, with ordinary pairs spread evenly. Nobody was examining that link. We pushed all 2,653 comparable pairs through the same ensemble, treated corridors included, which is conservative. Under a calibrated null, their percentiles come out uniform. They do not. Roughly 28% land below the 5th percentile and 27% above the 95th, against 5% expected at each tail. The bands run approximately 4.1× too tight, because the model jitters weights around an initialized value while real pairs move for reasons it does not carry. The model is not biased in location; it is understated in scale, and every percentile it reports is computed against a distribution about a quarter as wide as the world it scores. A 12th percentile is therefore unremarkable (Appendix B3).

The repair is Efron's. Build the null from the observed spread of all deviations rather than the model's own bands, replacing a theoretical null with an empirical one. Ranked that way, China–US sits near the 34th percentile and Vietnam–US near the 75th. Direction survives. Magnitude is smaller. A 2017–2019 backtest puts China–US near the 22nd percentile in 2019, so the instrument is reading a shock already on the record (Appendix B3).

§6. Why this is a paradigm change and not a cycle

GDP generated the network's topology through 2020. After that the weights and the deletion pattern leave it. The residual is not noise. It concentrates in the largest links and the named corridors, exactly where policy contests, and a structured residual against a fixed generator is the signature of a changed generating process, not of a model that was always approximate. Stated plainly, the model still gets the ordinary parts of the map right. It misses in one place, the place policy aims. A model that misses only there has not failed. It has dated a change.

The generator worked because those twenty-five years had one objective. Every node between 1996 and 2020 solved the same problem of minimizing cost, so economic mass predicted the topology. The model breaking after 2020 is not a failure. It is a paradigm ending, measured in dollars, on an instrument calibrated before it ended.

Bretton Woods giving way to floating rates in 1971 changed the price mechanism. The topology held; the same countries traded with the same countries at different rates. Post-Cold-War hyperglobalization changed the scale, adding nodes, edges, and depth to the cores, but the objective held too, because the entrants joined to minimize cost like everyone already inside. This transition changes neither price nor scale. It changes what the network optimizes, and that is the first such change since the Cold War.

The Cold War is the closest analog and a poor one. Those blocs were disconnected, so the objective could be met by separation, and once met it stayed met. These are entangled, which is why §4 measures re-routing. The system is not splitting in two; it is lengthening its paths to buy insurance and paying in efficiency, so the cost shows up as suppressed corridors and excess deletion, not a wall. That is also why the new objective does not settle. Cost was minimized against nature, which moves slowly, so the old program converged and stayed there. Dependency is minimized against an adversary who answers back, and every re-route makes a new chokepoint worth controlling.

The strongest cyclical rival is China's rebalancing, which suppresses the corridor and feeds the connectors without any flip elsewhere. What separates them is where the excess deletion sits.

Each earlier transition carried an event to date it, whether the gold window closing, the Wall coming down, or WTO accession. This one has none, which is why it shows only as a residual against a model that ended.

Any framework that mean-reverts to the pre-2020 network is mispricing it, because the target has changed, not merely moved. Our falsifier inverts that. If the aggregate divergence index of Appendix B2 flattens through 2028 with tariffs still in place, this was a cycle, and we were wrong.

Part 3: Analytical Appendix

A. Methods, data, and protocol

Every series, model, and rule is named here before Parts 1 and 2 use them.

Data. CEPII BACI HS17 V202601 (2017–2024), the standard product-level bilateral trade database [28]; World Bank World Development Indicators (WDI), the fitness input; IMF World Economic Outlook (WEO) April 2026 (GDP paths 2025–2028); US Census FT900 partner imports; SEC EDGAR filings; EIA Monthly Energy Review Table 7.2a; Bureau of Labor Statistics (BLS) Current Population Survey (CPS) series LNU04034021 and LNU04027662; the Federal Register application programming interface for export-control rules; Lawrence Berkeley National Laboratory (LBNL) data-center and Queued Up series; the Edison Electric Institute large-load tariff tracker; Center for a New American Security (CNAS) Entity-List tallies. All trade values are nominal current dollars, the gravity fit runs on directed flows where the generator runs on undirected pairs, and GDP enters at one vintage; each is a known margin we have not corrected for.

Models. We employ the published synthetic world-trade-web model [2], which is aggregate, undirected, and GDP-driven. That scope forces the no-rewiring counterfactual and attack engine (B2). Chokepoints come from the BACI six-digit Harmonized System (HS6) product-level multigraph (B1), and B3 carries the learned checks.

Resolution. For A2 we count named China-based entities in BIS final rules citing semiconductor, advanced-computing, or AI grounds, omitting addresses and non-China entries. Every forecast resolves on an already-published fixed schedule. Candidates needing bespoke construction or a disclosure fallback were cut rather than printed with a soft resolver.

A1. USITC DataWeb annual figures, calculated duties over total customs value of goods imports; the collected rate, not the statutory schedule, and the two diverge. Chronology behind the A1 claim: the Supreme Court struck the IEEPA tariff authority on February 20, 2026, and the wall was rebuilt under Section 122 within four days; when Section 122's 150-day clock lapsed on July 24 by operation of law, a Section 301 replacement covering 60 economies on a two-tier 10/12.5% structure took effect the same minute. Two challenges sit before the Court of International Trade with no injunction issued, and the Federal Circuit's stay keeps the Section 122 appeal alive. **A2.** Federal Register final rules, counting convention above. **A3.** An enforcement announcement or documented license action by MOFCOM, China's commerce ministry; the current suspension ends November 10, 2026. The instrument behind the claim is China's unified supply-chain-security framework, State Council Order No. 834 of March 31, 2026. **B4.** US Census annual partner-level data, general imports; ASEAN is the ten members, Vietnam, Thailand, Malaysia, Indonesia, the Philippines, Singapore, Cambodia, Laos, Brunei, and Myanmar. **B5.** NVIDIA's FY2028 annual report, geographic-segment note, revenue by customer billing location; FY2026 was 9.1%. **C6.** Annual and quarterly SEC filings including finance-lease additions, Microsoft calendarized. **C7.** The LBNL data-center series over EIA total net generation; LBNL reports consumption and EIA reports generation, a denominator roughly 5% larger, and the threshold is set on the generation basis deliberately. If LBNL publishes no calendar-2028 actual by June 30, 2029, the EIA Electric Power Annual data-center class settles it. **C8.** The Edison Electric Institute large-load tariff tracker as primary, counting states with a utility-commission final order approving a large-load or data-center rate class; state commission dockets settle ties. 24 approved and 6 pending as of July 2026. **C9.** Quarterly SEC filings, same basis as C6. **D10.** BLS Current Population Survey series LNU04034021 against LNU04027662, not seasonally adjusted, each quarter the simple average of its three monthly values; the gap stood at +0.53pp in 2026Q2.

Protocol. We employed AI assistants for data gathering, literature retrieval, and adversarial review. We designed and ran every model and set every probability ourselves, after a verification pass held separate from model construction, per the forecasting literature [29–31].

B. The network models

We build a sectoral multigraph and a ported generator with a counterfactual, then checks that try to break both.

B1. Dependency multigraph. An aggregate generator cannot see a sector, so B1 serves the chokepoint half. We build directed product-layer graphs on BACI HS6 (2024). The layers are semiconductors with their manufacturing equipment, critical minerals and rare earths, pharmaceutical ingredients, and batteries, with a technology overlay running across them. Edge weight is export value and betweenness distance is 1/ln(1 + w). Edges below $1,000 are omitted. Concentration is the Herfindahl–Hirschman index on the 0–10,000 scale, where higher means more concentrated. Taiwan arrives in BACI as code 490, "Other Asia, nes," and is recoded to TWN before any centrality claim. The five tabulated products are chosen for single-origin concentration on goods with no short-run substitute, not for trade value.

The layered view is what exposes the asymmetry behind B5. Taiwan is the semiconductor layer's largest origin at roughly 19.6% of $1.19T. It ranks 2nd by eigenvector centrality but only 19th by betweenness, making it an origin rather than a broker, with approximately 43% of its chip exports going to China and Hong Kong. Taiwan can be severed but cannot sever.

Chokepoint product (HS6)	Origin	Top share (%)	2024 trade ($ B)
NdFeB sintered magnets (850511)	China	62.7	5.0
Li-ion cells and packs (850760)	China	58.9	115.7
Memory ICs (854232)	South Korea	37.5	202.3
Wafer-fab equipment, including EUV (848620)	Netherlands	25.7	77.5
Logic ICs (854231)	Taiwan	20.6	389.4

Table B1. A single origin holds roughly 21–63% of world exports in each named chokepoint product (BACI HS6, 2024).

Share and stakes run opposite. Magnets hold 62.7% on about $5.0B of trade; logic chips 20.6% on $389.4B. A ranking on trade value alone misses the tightest constraint.

B2. The inherited generator and what we add. The actual network can only be scored against a null. We port the synthetic world-trade-web model of Kennedy et al. [2], on which one of us is a co-author, Equations (2)–(8), to Python with three documented implementation choices: a floor of 0.05 on the Equation 8 adjustment factor, which the published paper leaves undefined; clamping the edge probability to [0, 1]; and single-Generator seeding. Neither touches the published functional forms. With w_i the GDP of country i across N countries, the fitness, edge-creation, and edge-weight laws are

$$x_i \equiv \frac{w_i}{\sum_{k=1}^{N} w_k / N}, \qquad P_L[x_i, x_j] = \frac{\alpha\, x_i x_j}{1 + \beta\, x_i x_j}, \qquad e_{ij} = 10^{-F} \cdot \min(w_i, w_j), \qquad F \sim \Gamma(6.5571,\ 0.5794)$$

with $(\alpha_0, \beta_0) = (220, 80)$ at initialization and $(\alpha, \beta) = (200, 80)$ in growth years. Existing edges are deleted as a function of relative weight,

$$y \equiv \log_{10}\!\left(\frac{e_{ij}}{w_i + w_j}\right), \qquad P_D[x_i, x_j] = \begin{cases} 10^{ay^2+by+c}, & y \geq -10, \\ 0.36, & \text{otherwise.} \end{cases}$$

with (a, b, c) = (−0.048303, −0.96089, −5.2225) fitted on 1996–2020 transitions. Surviving weights jitter with paired GDP growth per Equation (8).

Our additions start with initialization from the actual BACI 2020 network in place of a synthetic draw. GDP paths forward come from actual WDI and IMF WEO series. The ensemble runs 300 fixed-seed evolutions over 2021→2028, which we read as a null distribution and not as a replication target. Attack percolation and B3's checks sit on top.

All six printed deviations land under approximately 3%. The three dispersion statistics we omit are noisier. Clustering dispersion is the worst of them at roughly −10%, which still sits inside the published dispersion and points in no consistent direction. That leaves us an inherited model, not a re-tuned lookalike, and only an inherited model can have its ensemble read as a null.

The attack result rests on a mismatch between two distributions. In a network this dense, the giant component survives because connectivity is cheap and redundant, while value does not survive because it is concentrated in a heavy-tailed, log-gamma weight distribution. The equilibrium fragmentation costs in the literature [9] differ from this bound in units as well as horizon, gross flows against real income, and the bridge between

Statistic (avg 1996–2020)	Real network	Kennedy et al. [2] synth.	This port	Dev. vs synth.
Edges E	9,656.0	9,180.1	9,443.5	+2.9%
Density	0.370	0.368	0.362	−1.6%
Mean degree	119.205	113.335	116.586	+2.9%
Mean shortest path	1.260	1.294	1.276	−1.4%
Mean clustering coeff.	0.863	0.872	0.876	+0.4%
Max k-core	92.6	96.9	99.6	+2.8%

Table B2. Port validation: 30-iteration averages under the published bootstrap-GDP conditions against that paper's Tables 1–2. Endpoint years 1996 and 2020 agree within ±3.7%.

them is the substitution elasticity, near zero at impact and rising with time; a forecaster carrying only the equilibrium figure is surprised twice, once when the value strands and again when it returns. The bound is conservative in the other direction too, since removal here is static while cascading failure and input–output amplification [22, 23] widen the same gap, and endogenously formed networks sit near the critical point [24], so rewiring is the expected response rather than an anomaly.

Weights enter model units through a scale factor κ, from median-matching the fitted log-gamma law, roughly −0.30 to −0.36 dex by panel, against a median relative weight of approximately 3.61 in $-\log_{10}$. Bands reflect seed noise at point estimates, and the published fit reports no standard errors, so we do not propagate parameter uncertainty. B3 goes and measures that gap instead of assuming it away.

Aggregate divergence from the benchmark climbs every year from 2021 to 2024, standing between roughly 1.47× and 1.76× model noise across those years. §5 shows that scale is understated, so what we lean on is that it climbs, not how far. The suppressed-corridor index puts China–US approximately 14% below the counterfactual median by 2023–2024, at 0.86×, against a counterfactual path that keeps rising through 2028; Vietnam–US runs at about 1.26× (Exhibit X1-L). For every seven dollars GDP growth alone would carry, the China–US corridor now moves about six.

Because the ensemble starts at 2020, the 2018–2019 suppression already sits in its initial condition. The gravity benchmark is deliberately spare, a per-year least-squares fit of log flow on the two log GDPs alone, with no distance or agreement terms, so cross-pair levels flatter geographically close pairs, and we put our weight on the within-pair movement, 0.48× to 0.35×, which the spare specification holds fixed. It reads the larger total displacement, at 0.35× for China–US and 16.2× for Vietnam–US, and the generator reads the incremental movement since 2020. Excess deletion and connector gain complete the set, the latter visible in B4's gap of +11.6pp in 2023 rising to +25.7pp year-to-date 2026.

The attack engine runs strength-targeted removal percolation with 100-seed bands (Exhibit X1-R); policy enters as a parameter, with deletion-probability interventions on targeted pairs and tariffs as weight decay. That engine is the wrong instrument for a sector question. Asking an aggregate, undirected generator to price one product-level export control is a category error, which is why B1 sits beside it.

B3. Out-of-sample validation and learned checks (Exhibit X3). B3 tries to break the inherited laws rather than confirm them, taking the 1996–2020 fits to unseen data.

Weight law: on 14,308 real 2024 dyads, the weight distribution matches $\Gamma(6.5571, 0.5794)$ up to a single median shift of approximately −0.30 dex, the same κ we employ at initialization. Kolmogorov–Smirnov distance falls from roughly 0.090 to 0.022 after that shift. Out of sample the law survives, once scale is allowed for.

Deletion law: on 15,611 dyad transitions from 2020 to 2024, approximately 9.4% died on the cleaned panel, from which we omit known post-2022 non-reporters so that reporting mortality is not read as trade mortality. A random forest given no functional form recovers the quadratic's level and its 0.36 floor at the small-edge end. Substantial edges died at roughly 2.84% against the 0.50% the fitted law generates, an excess of approximately 6× that survives every control we applied. Six large edges die for every one the law expects. The excess shows up exactly where §1 says policy aims, at links too large to die by chance.

Attribution: a forest regressing each pair's 2017→2024 share change on scale and exposure features finds gravity dominant. GDP product and growth carry the importance mass, against an out-of-bag R^2 of only 0.07. Most pair-level churn is idiosyncratic. Strategic-layer mass alone ranks with gravity. Treated edges are too few to cross-section, at one China–US pair in 5,996, so corridor claims run through the ensemble rather than the regression.

Calibration record, the numbers behind §5. The tails are 28.3% below the ensemble's 5th percentile and 27.5% above its 95th, at a Kolmogorov–Smirnov distance to uniform of 0.239; the generator's within-pair dispersion is 0.104 dex against a 0.421-dex real cross-section, where one dex is a factor of ten, so the over-dispersion of approximately 4.1 in standard deviation is roughly 17 in variance. Uniformity placebo, 2020→2024: roughly 2,653 comparable pairs; KS distance to uniform 0.239; approximately 28.3% below the ensemble's 5th percentile and 27.5% above its 95th; cross-sectional standard deviation 0.421 dex against a median within-dyad model figure of 0.104 dex, an over-dispersion of roughly 4.1×. Pre-period 2017→2019: n = 2,928, KS 0.305, tails 34.7% and 30.8%, over-dispersion 10.7×. Bands roughly four times too narrow have no business carrying a p-value. That is why §5 prints ranks. Empirical null, ranked cross-sectionally: China–US 0.86× at the 34th percentile; Mexico–US 0.94× at the 43rd; China–Vietnam 1.25× at the 75th; Vietnam–US 1.26× at the 75th; pre-period China–US 0.78× at the 22nd. Scripts `71_placebo.py`, `72_empirical_null.py`, `73_calib_export.py`, fixed seeds, reproducible. The test has a boundary worth naming: it verifies the calibration of the bands, not the correctness of the mechanism, so a generator wrong in structure could still pass it. Ours failed on calibration, which is why the corridor claim stands as a rank against real country pairs rather than a p-value it never earned. Two extra ensemble runs bought the separation between a percentile that sounds like evidence and one that is. An ensemble that has not been placebo-tested is not a null but a distribution with a confident shape, and confidence is the one thing a null may not supply. We would apply the same test to any ensemble whose percentiles are quoted as evidence, including the ensemble-supervisor-calibration doctrine we otherwise follow [36].

Only one of the three panels breaks, and it breaks at the top of the size distribution. We therefore route corridor claims through the ensemble and sector claims through B1.

C. Re-optimization and shadow prices

Two optimization programs sit on the network and turn it into rankable policy pressure. One is minimum-dependency sourcing over the B1 chokepoint sectors. The other allocates capex against power for the AI buildout. Both are solved in Pyomo. HiGHS handles the linear programs, and multipliers for the nonlinear form come from a trust-region NLP. Duality follows Boyd and Vandenberghe [32].

The dual values are the payload. A binding constraint's shadow price names the policy pressure that pays most to relieve, and that is what makes a dual an escalation predictor. Two are on the record so far. The power dual points at the large-load tariff wave, C8's mechanism, now at 24 states

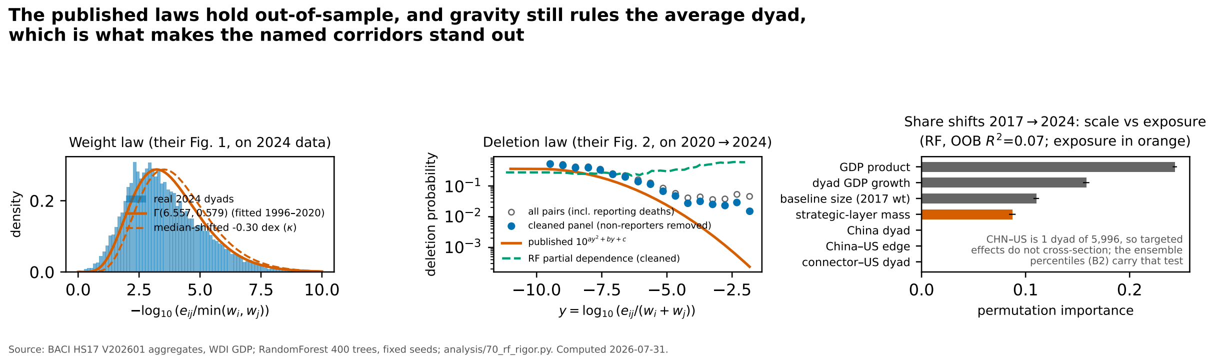


Exhibit X3. Out-of-sample fit curves and learned checks. The weight law is tested on 2024 data and the deletion law on 2020→2024 transitions, with share-shift attribution alongside.

approved plus 6 pending. The dependency dual names extraterritorial enforcement, which is A3. Where no actor can relieve a dual inside the window it names a pressure with no lever, so we read only the ones a regulator or a firm can actually move.

Exhibit X4 plots capex against the power envelope. Three series build that envelope. EIA net generation runs approximately 4,456 TWh over the trailing twelve months. The LBNL 2028 reference case is 464 TWh. Active interconnection-queue requests total roughly 2,061 GW, with a median request-to-operation wait above five years and only about 13% historical completion. Roughly seven of every eight queued gigawatts never energize.

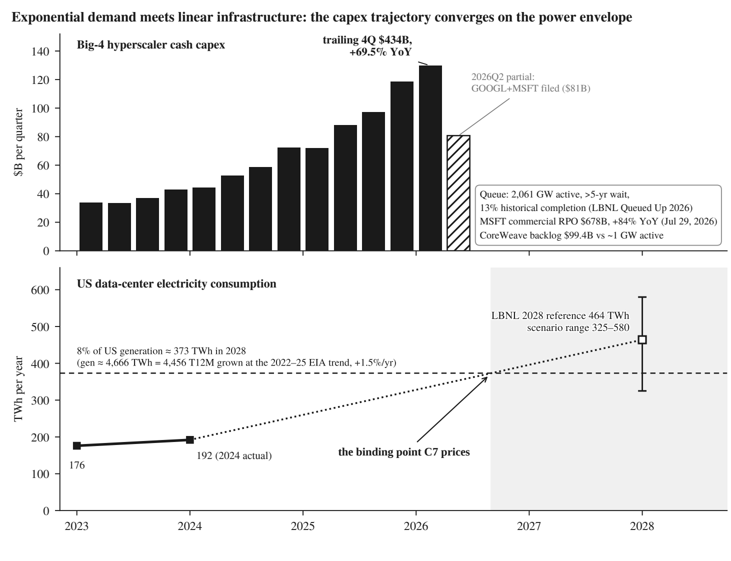


Exhibit X4. Capex trajectory against the power envelope, binding point annotated with its dual.

We price C7 through C9 against the crossing point rather than the trajectory. That dual is only as good as the tariff wave behind it. Appendix E states the falsifier that would retire it.

D. Monte Carlo calibration

Every printed probability is built the same way. It starts from an outside-view base rate, takes parameter distributions from the dated evidence files, and runs 40,000 Monte Carlo paths on fixed seeds. Nothing is extremized. The slate and its rationales are locked and dated. The Monte Carlo posteriors themselves are not redistributed.

C9 is the worked example. Combined Big-4 capex growth enters at a measured +69.5% over the latest four quarters, on a trailing-four-quarter base of $434B, with 2026Q1 at +80.5% year over year. On the basis question, entry rates come from the cash-capex series excluding finance leases, while C6 and C9 resolve on the lease-inclusive basis per Part 1. The walk-through runs on growth rates, which the basis shift moves far less than levels. Quarterly growth then follows a three-state process, the states being persist, constraint-bind and demand-crack. Growth decays roughly 14% per quarter toward a plateau of $q_\infty \approx 2\%$, at mean-reversion speed $\lambda \approx 0.22$, with the demand-crack state carrying about 15% weight. Year-over-year growth falls below 10% in two consecutive quarters before end-2028 on approximately 42% of paths, giving P(C9) = 0.42. The modal world compounds; a near-coin-flip minority cracks inside the window.

We place thresholds against measured values and re-place them when the measurement moves. We raised B4 to 30pp after the measured gap ran +25.7pp year-to-date 2026. We re-armed B5 at 12% after the FY2026 annual report showed a 9.1% China and Hong Kong revenue base with data-center compute at zero. We raised C7's threshold to 8% after LBNL's June 2026 update. We re-priced A2 from 50 to 38 when an audit found its base rate frozen: 95 China-based additions in 2025 on CNAS's count, then no entity-addition final rule on those grounds since October 9, 2025. That makes A2 a resumption bet on the truce decaying past the November suspension end-dates.

C6's 45 is priced rather than timid. The $725B of guided 2026 spending needs only approximately 38% further growth to clear $1.0 trillion in 2027. Depreciation drag on a fast-aging fleet bites inside the window. So do rising debt cost and base effects on a doubled denominator. That is the arithmetic behind C9's crack.

Where a printed number differs from the raw model draft (A1 62 against 64; C7 75 against 82; B5 20 against 12), the overlay is a documented judgment on risks the trees do not carry, among them litigation tempo, definitional and timing risk, and the licensed-upside tail. The C7 haircut prices the canonical warning that IT-load forecasts overshoot [33]. B5 carries the largest relative move of the three at approximately 67% above the model draft, and its dated rationale is filed in `notes/B5-OVERLAY-2026-08-01.md` in the replication repository. That overlay is the weakest seam in this paper, and endogenizing litigation tempo is the next piece of work. Calibration and sharpness follow the tournament literature [34, 35] and the ensemble-supervisor-calibration doctrine from Bridgewater's own forecasting lab [36], and sub-even calls stand where base rates disagree with narrative. Scored against our own probabilities, our expected Brier score is approximately 0.23.

E. Sensitivity and falsifiers

We state in advance what would make this slate wrong.

Sensitivities. Decomposing the excess deletion by dyad, which is what separates the rebalancing story from ours in §6, is the next piece of work. Section 301 litigation outcomes and any comprehensive truce move Group A. A Taiwan Strait event moves most of B and C down. A federal power fast-track moves C7 through C9 down. CPS revisions move D10. A further truce extension past the November 10 and November 27, 2026, suspension end-dates delays extraterritoriality and moves A3 down. One common shock sits behind all of them. A Treasury-demand break of the late-big-cycle type [37] would move all four groups at once, not one at a time.

Acemoglu [19] is the counterweight on labor magnitudes. Occupational-exposure estimates run considerably larger [38]. D10 sits between the two.

Falsifiers. Should C8 stall below 40 states, the Appendix C model overweights the power dual. A YES on B5 at 12% would mean bilateral lock-in is softer than we claim. And if the Section 301 regime is enjoined while calendar-2027 collections still exceed 8%, the channel-switching mechanism is stronger than we priced. The slate-level falsifier stands above them all: if the Appendix B2 divergence index flattens through 2028 with tariffs still in place, this was a cycle and we were wrong.

F. References

[1] Greg Jensen. Two Resource Grabs Reshaping Markets: Modern Mercantilism and AI. Bridgewater Associates, recorded remarks, 2026.

[2] Sean Kennedy, Michael Wish, Philip Smith, James Sherrell, William Shields, and Ralucca Gera. Building a reliable, dynamic and temporal synthetic model of the World Trade Web. In Complex Networks XIII (CompleNet 2022), Springer Proceedings in Complexity. Springer, 2022.

[3] M. Ángeles Serrano and Marián Boguñá. Topology of the World Trade Web. Physical Review E, 68:015101(R), 2003.

[4] Diego Garlaschelli and Maria I. Loffredo. Structure and evolution of the world trade network. Physica A, 355(1):138–144, 2005.

[5] Assaf Almog, Tiziano Squartini, and Diego Garlaschelli. A GDP-driven model for the binary and weighted structure of the International Trade Network. New Journal of Physics, 17:013009, 2015.

[6] Karen Karniol-Tambour. Taking Stock of the New Paradigm. Bridgewater Associates, July 2026.

[7] Nir Bar Dea. On the new world order, AI, and Israel's risks. Interview, Unholy: Two Jews on the News, 2026.

[8] Henry Farrell and Abraham L. Newman. Weaponized interdependence: How global economic networks shape state coercion. International Security, 44(1):42–79, 2019.

[9] Shekhar Aiyar, Jiaqian Chen, Christian Ebeke, et al. Geoeconomic fragmentation and the future of multilateralism. Technical Report Staff Discussion Note SDN/2023/001, International Monetary Fund, 2023.

[10] Gita Gopinath, Pierre-Olivier Gourinchas, Andrea F. Presbitero, and Petia Topalova. Changing global linkages: A new Cold War? Journal of International Economics, 2025.

[11] Greg Jensen. We're All Mercantilists Now. Bridgewater Daily Observations, November 2024.

[12] Greg Jensen. Understanding AI's More Dangerous Phase. Bridgewater Associates, February 2026.

[13] Greg Jensen et al. The Macro Implications of the AI Capex Boom. Bridgewater Daily Observations, January 2026.

[14] Jaime Sevilla and Edu Roldán. Training compute of frontier AI models grows by 4–5x per year. Epoch AI, May 2024.

[15] Josh You and David Owen. How much power will frontier AI training demand in 2030? Epoch AI with EPRI, August 2025.

[16] Arman Shehabi, Sarah J. Smith, Alex Hubbard, Alexander Newkirk, Nuoa Lei, et al. 2024 United States data center energy usage report. Technical Report LBNL-2001637, Lawrence Berkeley National Laboratory, December 2024.

[17] Sarah J. Smith, Alex Hubbard, Alexander Newkirk, et al. United States data center energy usage report: 2025 update. Technical report, Lawrence Berkeley National Laboratory, June 2026.

[18] Karen Karniol-Tambour et al. Power Politics: Energy Self-Sufficiency in a Modern Mercantilist World. Bridgewater Associates, October 2025.

[19] Daron Acemoglu. The simple macroeconomics of AI. Economic Policy, 39(120):813–861, 2024.

[20] Daron Acemoglu, Vasco M. Carvalho, Asuman Ozdaglar, and Alireza Tahbaz-Salehi. The network origins of aggregate fluctuations. Econometrica, 80(5):1977–2016, 2012.

[21] Daron Acemoglu, Asuman Ozdaglar, and Alireza Tahbaz-Salehi. Systemic risk and stability in financial networks. American Economic Review, 105(2):564–608, 2015.

[22] David Baqaee and Emmanuel Farhi. Networks, barriers, and trade. Econometrica, 92(2):505–541, 2024.

[23] Alexandr Kopytov, Bineet Mishra, Kristoffer Nimark, and Mathieu Taschereau-Dumouchel. Endogenous production networks under supply chain uncertainty. Econometrica, 92(5):1621–1659, 2024.

[24] Matthew Elliott, Benjamin Golub, and Matthew V. Leduc. Supply network formation and fragility. American Economic Review, 112(8):2701–2747, 2022.

[25] Caroline Freund, Aaditya Mattoo, Alen Mulabdic, and Michele Ruta. Is US trade policy reshaping global supply chains? Journal of International Economics, 152, 2024.

[26] Laura Alfaro and Davin Chor. Global supply chains: The looming "great reallocation". Technical Report Working Paper 31661, NBER, 2023.

[27] Richard Baldwin, Rebecca Freeman, and Angelos Theodorakopoulos. Hidden exposure: Measuring US supply chain reliance. Technical Report Working Paper 31820, NBER, 2023.

[28] Guillaume Gaulier and Soledad Zignago. BACI: International trade database at the product-level. Technical Report Working Paper 2010-23, CEPII, 2010.

[29] Danny Halawi, Fred Zhang, Chen Yueh-Han, and Jacob Steinhardt. Approaching human-level forecasting with language models. arXiv:2402.18563, 2024.

[30] Philipp Schoenegger, Peter S. Park, Ezra Karger, Sean Trott, and Philip E. Tetlock. AI-augmented predictions: LLM assistants improve human forecasting accuracy. arXiv:2402.07862, 2024.

[31] Ezra Karger, Houtan Bastani, Chen Yueh-Han, Zachary Jacobs, Danny Halawi, Fred Zhang, and Philip E. Tetlock. ForecastBench: A dynamic benchmark of AI forecasting capabilities. arXiv:2409.19839, 2024.

[32] Stephen Boyd and Lieven Vandenberghe. Convex Optimization. Cambridge University Press, 2004.

[33] Jonathan Koomey and Eric Masanet. Does not compute: Avoiding pitfalls assessing the internet's energy and carbon impacts. Joule, 5(7):1625–1628, 2021.

[34] Philip E. Tetlock and Dan Gardner. Superforecasting: The Art and Science of Prediction. Crown, 2015.

[35] Barbara Mellers, Lyle Ungar, Jonathan Baron, et al. Psychological strategies for winning a geopolitical forecasting tournament. Psychological Science, 25(5):1106–1115, 2014.

[36] Rohan Alur, Bradly C. Stadie, Daniel Kang, Ryan Chen, Matt McManus, Michael Rickert, Tyler Lee, Michael Federici, Richard Zhu, Dennis Fogerty, Hayley Williamson, Nina Lozinski, Aaron Linsky, and Jasjeet S. Sekhon. AIA Forecaster: Technical report. arXiv:2511.07678; Bridgewater AIA Labs, 2025.

[37] Ray Dalio. Principles for Dealing with the Changing World Order. Avid Reader Press / Simon & Schuster, 2021.

[38] Tyna Eloundou, Sam Manning, Pamela Mishkin, and Daniel Rock. GPTs are GPTs: An early look at the labor market impact potential of large language models. arXiv:2303.10130, 2023.